

<i>Power Station</i>	<i>Location</i>	<i>Power Station</i>	<i>Location</i>
Tarapur	Maharashtra	Rawatbhata	Rajasthan
Kalpakkam	Tamil Nadu	Narora	Uttar Pradesh
Kakrapara	Gujarat	Kaiga	Karnataka
Kudankulam	Tamil Nadu	Banswara	Rajasthan

(UMPP)

<i>Plants</i>	<i>States</i>	<i>Capacity (MW)</i>	<i>Awarded to</i>
Sasan	Madhya Pradesh	4000	Reliance
Mundra	Gujarat	4000	Tata
Krishnapatnam	Andhra Pradesh	4000	Reliance
Girye	Maharashtra	4000	NA
Tadri	Karnataka	4000	NA

<i>States</i>	<i>Name of the Plant</i>
Haryana	Faridabad, Panipat
Punjab	Bathinda, Ropar
Delhi	Badarpur, Indraprastha
Rajasthan	Kota
Uttar Pradesh	Obra, Panki, Singrauli
Gujarat	Ukai, Sikka, Ahmedabad Sabarmati
Madhya Pradesh	Satpura, Amarkantak, Pench
Chhattisgarh	Korba, Bhilai
Maharashtra	Nashik, Uran, Chandrapur Trombay, Dabhol
Andhra Pradesh	Ramagundam, Kotha-gundam, Nellore, Vijayawada
Tamil Nadu	Ennore, Tuticorin, Neyveli
Bihar	Barauni
Jharkhand	Bokaro
Odisha	Talcher, Rourkela
West Bengal	Kolkata, Titagarh, Durgapur
Assam	Namrup, Bongaigaon
Jammu & Kashmir	Pampore
Tripura	Rokhia

Non-Conventional or Renewable Energy Sources

- The non-conventional energy sources include solar energy, wind energy, biomass energy, fuel cell, electric vehicles, tidal energy, hydrogen energy and geothermal energy. The renewable energy programme started with the establishment of the Department of Non-Conventional Energy Sources in 1982 in India. Indian Renewable Energy Development Agency was set-up in 1987.

- In 1992, DNES was converted into Ministry of Non-Conventional Energy Sources, which is renamed in 2006 as Minister of New and Renewable Energy (MNRE). The minister now has taken up some programmes on various new technologies.

Renewable Energy Plants

<i>Types of Energy</i>	<i>Plants</i>	<i>States</i>
Wind energy	Muppandal Perungudi	Tamil Nadu
	Kayattar	Tamil Nadu
	Satara	Maharashtra
	Jogimati	Karnataka
Geothermal energy	Lamba Mandvi	Gujarat
	Manikaran	Himachal Pradesh
	Puga Valley	Jammu and Kashmir
	Tattapani	Chhattisgarh
Tidal energy	Gulf of Khambat	Gujarat
	Gulf of Kachchh	Gujarat
	Sunderban	West Bengal
Wave energy	Vizhinjam	Kerala
Solar energy	Tirupati	Andhra Pradesh

Economic Activities

Human activities which generate income are known as economic activities. Economic activities are broadly grouped into primary, secondary, tertiary, quaternary and quinary activities.

- Primary Activities** are directly dependent on environment as these refer to utilisation of Earth's surface such as land, water, vegetation, minerals etc. Thus, includes hunting and gathering, pastoral activities, fishing, forestry, agriculture, mining and quarrying.
- Secondary Activities** add value to natural resources by transforming raw materials into valuable products.
- Tertiary Activities** include all types of services that required special skills provided in exchange of payments. These are health, education, law, governance etc.
- Quaternary Activities** centre around research, development and may be seen as an advanced form of services involving specialised knowledge and technical skills.
- Quinary Activities** are services that focus on the creation, re-arrangement and interpretation of new and existing technologies.

INDUSTRIES

The industries sector is regarded as the growth engine for economic development of a nation. As India is an emerging economy, in the post reform era 22% of the employment generation has been attributed to industrial sector.

<i>Industries</i>	<i>Details</i>
Cotton Textile Industry	The first modern cotton textile mill was established in Bombay in 1854 by local parsi entrepreneurs with the name of Bombay spinning and weaving company. Mumbai is called cottonopolis of India and Ahmedabad is called Manchester of India. Coimbatore is called Manchester of South India and Kanpur is called Manchester of Uttar Pradesh. Distribution Maharashtra (Mumbai, Sholapur, Pune, Kolhapur, Satara, Wardha, Aurangabad and Amravati), Gujarat (Ahmedabad, Vadodara, Rajkot, Surat, Bhavnagar, Porbandar, Maurvi and Viramgam), Tamil Nadu (Chennai, Tirunelveli, Madurai, Tuticorin, Salem, Virudhnagar and Pollachi), Karnataka (Bengaluru, Belgaum, Mangaluru, Chitradurga, Gulbaraga and Mysore), Uttar Pradesh (Kanpur, Etawah, Modinagar, Moradabad, Bareilly, Agra, Meerut and Varanasi), Madhya Pradesh (Indore, Gwalior, Ujjain and Bhopal), Rajasthan (Kota, Jaipur, Sri Ganganagar, Bhilwara and Udaipur).
Woollen Textile Industry	The first woollen textiles mill was set-up in 1876 at Kanpur. Jammu and Kashmir is a large producer of handloom woollen goods. Distribution Punjab (Dhariwal, Amritsar, Ludhiana, Ferozpur), Maharashtra (Mumbai), Uttar Pradesh (Kanpur, Mirzapur, Agra and Tanakpur)
Jute Textile Industry	First modern jute mill was set-up in 1855 at Rishra near Kolkata. India is the largest producer of raw jute and jute good production, whereas it is second largest exporter of jute goods after Bangladesh. Distribution West Bengal, Bihar, Uttar Pradesh, Andhra Pradesh, Assam, Odisha, Tripura and Chhattisgarh.
Silk Textile Industry	India is the second largest producer of natural silk, after China and is the only country producing all four varieties or natural silk viz Mulberry, Tasar, Eri and Muga of which golden yellow Muga silk is unique in India. Distribution Karnataka is the leading producer followed by West Bengal, Bihar etc.
Rubber Industry	The first factory of synthetic rubber was set-up at Bareilly. Distribution Bareilly (Uttar Pradesh), Baroda (Gujarat) Synthetic rubber units, Mumbai, Ahmedabad, Amritsar-Reclaimed rubber units.
Tea Industry	Tea cultivation in India was first started in the mid-19th century in Darjeeling, Assam and Nilgiris. Nearly 98% of the tea production comes from Assam, West Bengal, Tamil Nadu and Kerala, while the rest of it comes from Karnataka, Terai region of Uttarakhand, Himachal Pradesh, Arunachal Pradesh, Manipur and Tripura.
Sugar Industry	Uttar Pradesh is the leading producer of sugar. Distribution Uttar Pradesh (Gorakhpur, Deoria, Basti, Gonda, Meerut, Saharanpur, Muzaffarnagar, Bijnor and Moradabad), Bihar (Darbhanga, Saran, Champaran and Muzaffarpur), Punjab (Phagwara and Dhuri), Haryana (Ambala, Rohtak and Panipat), Maharashtra (Nashik, Pune, Satara, Sangli, Kolhapur and Solapur) and Karnataka (Munirabad, Shimoga and Mandya).
Paper Industry	The first paper mill in the country was set-up at Serampore (Bengal) in 1832, which failed. In 1870, a fresh venture was started at Ballygunge near Kolkata. Raw material Bamboo (70%), Salai wood (12%), Sabai (9%), Bagasses (4%) and Waste paper and Rags (5%). Distribution Madhya Pradesh (Nepanagar), Hindustan Paper Corp, Vellore, Mysore Paper mill, Bhadravati, Maharashtra, (Mumbai, Pune, Ballarpur and Kamptee produce paper and vikhroli), Andhra Pradesh (Rajahmundry and Sirpur), Madhya Pradesh (Indore, Bhopal and Shandol) and Karnataka.
Iron and Steel	Distribution Bhadravati (Karnataka), Jamshedpur (Jharkhand), Durgapur, Burnpur (West Bengal), Bokaro (Jharkhand, Bhadravati) (Karnataka), Rourkela (Odisha), Bhilai (Chhattisgarh), Salem (Tamil Nadu) and Visakhapatnam (Andhra Pradesh).
Ship	Distribution Cochin Shipyard, Mumbai (Mazgaon Dock), Hindustan Shipyard at Visakhapatnam and Kolkata (Gorden Reach workshop). For Indian Navy, only at Mazgaon.
Aircraft Industry	Distribution Hindustan Aeronautics India Limited was formed by merging two aircraft factories at Bengaluru and Kanpur. Four other factories are at Nashik, Lucknow, Koraput (Odisha) and Hyderabad.
Fertilizer Industry	The Fertilizer Corporation of India (FCI) was set-up in 1961 and National Fertilizer Limited (NFL) was set-up in 1974. Distribution Sindri (Bihar), Nangal, Trombay, Gorakhpur (Uttar Pradesh), Durgapur, Namrup, Cochin, Rourkela, Neyveli, Varanasi, Vadodra, Kanpur, Visakhapatnam and Kota.
Heavy Machinery	Distribution Durgapur, Mumbai, Ranchi, Visakhapatnam, Tiruchirapalli and Naini.
Machine Tool Industry	It forms the basis for the manufacturing of industrial, defence equipments, automobiles, railway engines and electrical machinery. Distribution Hyderabad, Bengaluru, Pinjore (Haryana), Kalamassery (Kerala), Secunderabad, Ajmer and Srinagar.
Heavy Electrical Equipments	Distribution Bengalure, Bhopal, Jammu, Tiruchirapalli, Ramchandrapuram (Hyderabad) and Jagdishpur (Uttar Pradesh).
Photo Films Industry	The Hindustan Photo Films Manufacturing Company at Udagamandalam (Tamil Nadu) is the only factory in the public sector, producing photo paper and films.
Glass Industry	Distribution Uttar Pradesh (Firozabad, Bahjoi, Hathras, Naini, Shikandrabad), Maharashtra (Mumbai, Telogaon, Pune, Sitarampur), Tamil Nadu (Tiruvottiyor) and Karnataka (Bolgaon, Bengaluru).

TRANSPORT

Railways

- India has the fifth largest railway network in the World after the USA, Russia, China and Canada. The Indian railway operate in three different gauges

Gauge	Routes (km)
Broad Gauges (1. 676 m)	55000
Meter Gauges (1. 000 m)	~6809
Narrow Gauges (0.761 and 0.610 m)	~2463

- It is the largest public sector undertaking of the country and it is the world's largest railway network under single management.
- The first Indian railway line in India was operated for public traffic in 1853 between Mumbai and Thane over a distance of 34 km.
- The first electric train in India was 'Deccan Queen', it was introduced in 1929 between Bombay and Poona.
- The headquarters of Indian railway is in New Delhi.
- The fastest train in India is the Gatimaan Express, whose maximum speed is 160 km/hr.

Zone	Headquarter
Central railway	Mumbai
Eastern railway	Kolkata
Northern railway	New Delhi
North-Eastern railway	Gorakhpur
North-East Frontier railway	Malegaon
Southern railway	Chennai
South Central railway	Secundrabad
South-Eastern railway	Kolkata
Western railway	Mumbai (Church Gate)
East Central railway	Hajipur
East Coast railway	Bhubaneshwar
North Central railway	Allahabad
North-Western railway	Jaipur
South-East Central railway	Bilaspur
South-West railway	Hubli
West Central railway	Jabalpur

- Indian railways has the second biggest electrified system in the world after Russia.
- The total route covered is approximately 63000 km.
- Vivek Express has the longest train route in India connecting Dibrugarh and Kanyakumari. It is 8th longest in the world.

- The second longest train route is of 'Himsagar Express' from Jammu Tavi to Kanyakumari. It covers a distance of 3726 km and passes through ten states.
- The oldest steam engine 'Fairy Queen' still runs on rail.
- Uttar Pradesh has largest railway network in India.
- Mumbai CST is busiest railway junction of India.
- Railway track electrification was introduced in early 1920s. The first two sections from Victoria Terminus to Kurla and from Victoria Terminus to Bandra were electrified. About 26% of the rail lines have been electrified.
- Anil Kakodkar Committee was constituted for Rail Safety in 2011.

Railways	Specialities
Darjeeling Himalayan Railways (1999)	Narrow gauge railway from Siliguri to Darjeeling in the state of West Bengal
Chhatrapati Shivaji Terminus (2004)	It was opened in 1887, in the time to celebrate Queen Victoria's golden jubilee.
Nilgiri Mountain Railways (2005)	It connects the town of Mettupalayam with the hill station of Udagamandalam in the Nilgiri hills.
Kalka-Shimla Railways (2008)	Narrow Gauge railway in North-West India travelling along a most mountainous route from Kalka to Shimla.

*The year in bracket represents the year in which UNESCO has added the railway line to the World Heritage site list.

Konkan Railways

- It runs from Mangalore to Roha (40 km South of Mumbai).
- It has the fastest track in India and has a total length of 738 km.
- Almost 10% of the line passes through tunnels.
- Konkan railway connects Maharashtra, Karnataka and Goa.

City	System	Start of Operations	Notes
Kolkata	Kolkata Metro	24th October, 1984	First mass rapid transit system in India and the 17th zone of the Indian Railways.
Chennai	Chennai MRTS	1st November, 1995	It is planned for the MRTS to be taken over by the Chennai Metro Rail Limited once the Chennai Metro becomes operational.
Delhi	Delhi Metro	24th December, 2002	India's first modern rapid transit system.
Bengaluru	Namma Metro	20th October, 2011	First Metro in India to introduce Wi-Fi onboard trains.
Gurgaon	Rapid Metro Rail Gurgaon	14th November, 2013	India's first fully privately financed metro, and the first metro system in the country to auction naming rights for its stations.
Mumbai	Mumbai Metro	8th June, 2014	The largest number of underground stations (86 spanning 3 planned lines) in any metro system in India.

Roadways

In 1943, Nagpur plan classified the roads into four categories

- i. National highway
 - ii. State highway
 - iii. District roads
 - iv. Village roads
- Indian road network is the 2nd largest in the world.
 - India has a road network of over 4.8 million km.
 - National highways are constructed and maintained by Central Public Works Department (CPWD).

National Highway	Route	Distance
NH-1	New Delhi-Ambala-Jalandhar-Amritsar	456
NH-1 A	Jalandhar-Uri	663
NH-2	Delhi-Mathura-Agra-Kanpur-Allahabad-Varanasi-Kolkata	1465
NH-3	Agra-Gwalior-Nasik-Mumbai	1161
NH-4	Thane-Chennai via Pune and Bengaluru	1235
NH-5	Kolkata-Chennai	1533
NH-6	Kolkata-Dhule	1949
NH-7	Varanasi-Kanyakumari	2369
NH-8	Delhi-Mumbai (via Jaipur, Vadodra and Ahmedabad)	1428
NH-9	Mumbai-Vijayawada	841
NH-10	Delhi-Fazilka	403
NH-15	Pathankot-Samakhiali (along great Indian desert)	1526
NH-17	Parvel-Edappally (along West coast)	1269
NH-22	Ambala-Shipki La (along Shipki La Gorge)	459
NH-31	Barhi-Guwahati (connecting East to North-East)	1125
NH-37	Pancharatna near Goalpara-Saikhoaghat (along Assam plain)	680
NH-44	Srinagar to Kanyakumari	630
NH-47A	Kundanoor and Willingdon-Island in Kochi Kerala	6
NH-49	Cochin-Dhanshkodi	440
NH-52	Daihatsu-Junction NH-47 (near Saikhowaghat)	850
NH-58	Delhi-Mana	538
NH-65	Ambala-Pali	690
NH-75	Gwalior-Ranchi	955
NH-220	Kollam (Quilon)-Teui (TN Border)	265

Expressway

- Expressway is a highway especially planned for high speed traffic, limited point of access or exist. Expressway are the highest class of roads in the Indian roads network. Indian roads has approx 942 km expressway.

- The Ministry of Road Transport and Highways is already in the process of preparing a draft for creation of a National Expressway Authority of India (NEAI) on the line NHAI.

Some of the Important Information Regarding the National Highways

- NH44, is the longest NH of India, running from Srinagar to Kanyakumari
- NH5 and NH17 run along the Eastern and the Western coast respectively.
- NH15 represents the border road in Rajasthan desert.
- NH47A is the shortest highway in the Indian highway network.
- Maximum length of highway is present in State of Uttar Pradesh.

National Highway Development Programme (NHDP)

National Highway Development Programme consists of following projects :

- The Golden Quadrilateral project involves connectivity of
 - Delhi to Kolkata (NH2)
 - Delhi to Mumbai (NH8, NH76 and NH79)
 - Mumbai to Chennai (NH4, NH7 and NH46)
 - Chennai to Kolkata (NH5, NH6 and NH60)
 Total length **5846 km**, out of which maximum length is in Andhra Pradesh (1016 km) followed by Uttar Pradesh (753 km).
- North-South and East-West corridors
 - NS corridor connects Srinagar to Kanyakumari.
 - EW corridor connects Porbandar (Gujarat) to Silchar (Assam).

Airways

- In 1935, the 'Tata Airlines' started its operation between Mumbai and Thiruvananthapuram and in 1937 between Mumbai and Delhi.
- In 1953, all the private airline companies were nationalised and Indian Airlines and Air India came into existence.
- International Airports Authority of India and National Airports Authority were merged on 1995 to form Airports Authority of India. The Authority manages the Civil Aviation Training College at Allahabad and National Institute of Aviation Management and Research at Delhi.

International Airport	City
Rajiv Gandhi International Airport	Hyderabad
Calicut International Airport	Calicut
Chhatrapati Shivaji International Airport	Mumbai
Kempe Gowda International Airport	Bengaluru
Goa International Airport, Dabolim	Goa
Netaji Subhash Chandra Bose International Airport	Kolkata
Thiruvananthapuram International Airport	Thiruvananthapuram
Lokpriya Gopinath Bordoloi International Airport	Guwahati
Sardar Vallabhbhai Patel International Airport	Ahmedabad
Indira Gandhi International Airport	Delhi
Chennai International Airport	Chennai
Sri Guru Ram Dass Jee	Amritsar
Cochin International Airport	Cochin (Kerala)
Coimbatore International Airport	Coimbatore (Tamil Nadu)
Lal Bahadur Shastri Airport	Varanasi (Uttar Pradesh)
Chaudhary Charan Singh Airport	Lucknow (Uttar Pradesh)
Ambedkar Airport	Nagpur (Maharashtra)
Gaya Airport	Gaya (Bihar)
Imphal International Airport	Imphal (Manipur)

Waterways

Major Waterways of India

Numbers	Stretches of the Waterways	Specifications
NW1	Allahabad-Haldia (1620 km)	along Ganga river
NW2	Sadiya- Dhubri (891 km)	along Brahmaputra river
NW3	Kottapuram-Kollam (168 km)	along Champakara and Udyogmandal canal
NW4	Bhadrachalam to Rajahmundry and Wazirabad to Vijaywada (1095 km)	along Godavari and Krishna river
NW5	Mangalgarhi to Paradeep and Talcher to Dhamara (623 km)	along Mahanadi and Brahmini river system
NW6	Lakhipur to Bhanga (121 km)	along Barak river

Ports in India

- The waterways authority in India divides Indian ports into three categories, major, minor and intermediate.
- India has about 200 ports, with 13 major and the rest intermediate and minor.
- Project Sagarmala has been conceived for development.

Ports of Eastern Coast	Important Facts
Kolkata	Oldest port, India's revenue port having two dock system.
Paradip	It handles iron ore and some amounts of coal and dry cargo.
Chennai	All weather port having deep drafted berth, oil jetties, iron ore terminals, etc.
Visakhapatnam	Seaport and well known for its outstanding performance. It serves the Bhilai and Rourkela steel plant.
Tuticorin	Artificial deep sea harbour, all weather port offer direct weekly container service to USA.
Ennore	First corporatised major port in India.

Ports of Western Coast	Important Facts
Mumbai	It handles maximum traffic, natural harbour mostly petroleum and dry cargo.
Kandla	Tidal port and important traffic handled are crude oil, petroleum, edible oil, foodgrains.
Marmagao	It handles iron ore. It has a naval base.
New Mangaluru	It is an all weather port.
Cochin	Major natural port in Willingdon Island.
Jawaharlal Nehru	It is called as Nhava Sheva.

- Largest container port of India is Jawaharlal Nehru port in Mumbai. The largest natural port is in Visakhapatnam.
- Kandla in Gujarat is a tidal port. It has been made into a free trade zone.
- New Mangaluru port is also called the 'Gateway of Karnataka'.
- Mumbai port is the busiest port of India.
- Mundra port is largest private port of India.
- The Union Cabinet has given its in-principle approval for setting up a major port of Enayam near Colachel in Tamil Nadu. On completion the port will become country's 13th major port.

Demographic Profile of India

Population

- Population geography is closely related to demography. It is concerned with the study of demographic processes and their consequences in and environmental context.
- Population density is the degree of compaction in population the closeness of persons living on a given surface, the spatial balance of their social and economic assets.
- Population density shows the population pressure on land resource. There are various ways to measure population density such as crude or arithmetic density, nutritional or physiological density, agricultural density, economic density etc. Among them arithmetic density is mostly used.